

8 Scientists studied the distribution of biomass in organisms on Earth.

(a) State the meaning of the term **biomass**.

(1)

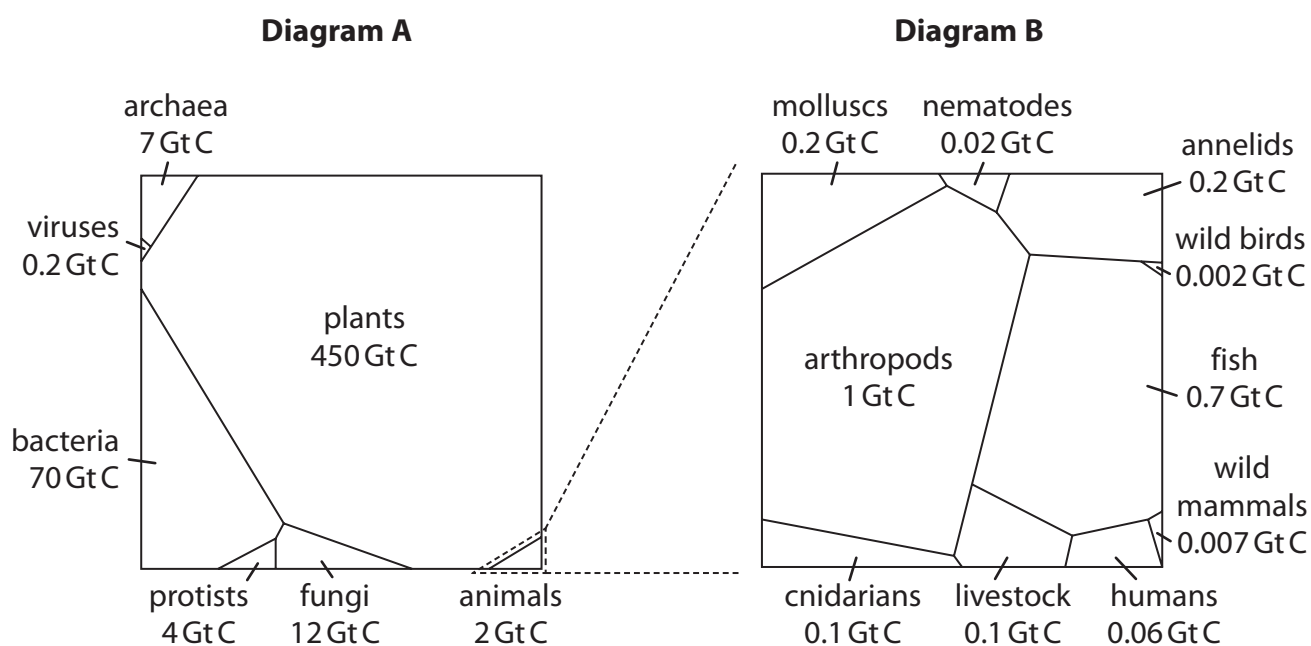
(b) Voronoi diagrams were used to present some of the data.

Two Voronoi diagrams are shown.

Diagram A shows the biomasses of groups of organisms.

Diagram B shows the biomasses of the organisms in the animals group.

The biomass is given in gigatons of carbon (GtC), where $1 \text{ GtC} = 10^{15} \text{ g}$ of carbon.



The area of each polygon is proportional to the biomass of that organism. The shape of each polygon has no meaning.

- (i) Calculate the percentage of biomass in organisms belonging to the domain Eukarya.

Use the information in diagram A.

(2)

Answer %

- (ii) Suggest why the scientists studied the distribution of biomass in groups of organisms and not the number of individual organisms.

Use the information in diagram A.

(2)

- (iii) Discuss the advantages and disadvantages of presenting data in Voronoi diagrams.

Use diagram B to support your answer.

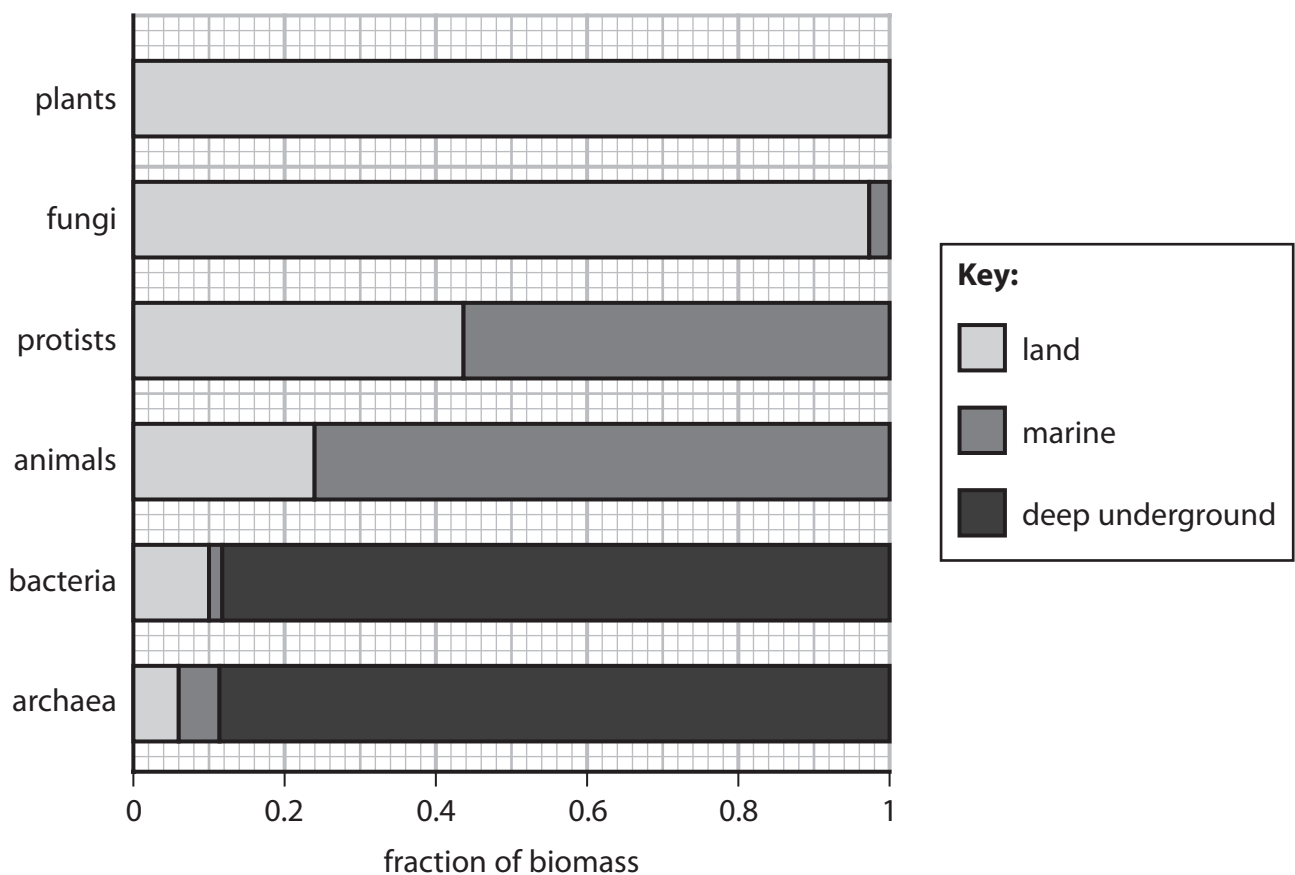
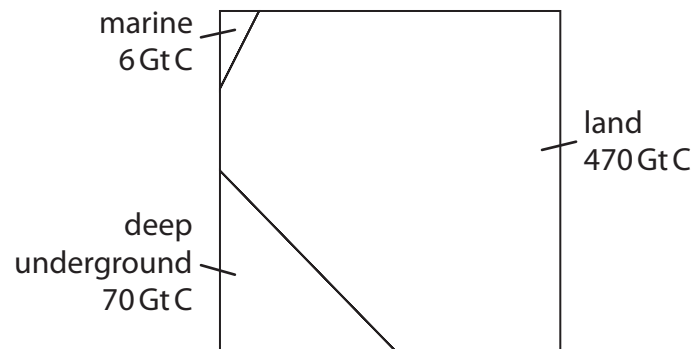
(3)

[illegible]

*(c) The same study also determined the distribution of biomass in three different environments: marine, deep underground and land.

The Voronoi diagram shows the distribution of biomass in each environment.

The graph shows the proportion of biomass in different groups of organisms.



Use the information in the Voronoi diagram and the graph to support your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

TOTAL FOR PAPER = 90 MARKS